

Education			
Year	Degree	Institution	Performance
2025 - Present	M.Sc. Statistics	Indian Institute of Technology Kanpur	8.66/10
2025	B.Sc. Statistics (Hons.)	The Maharaja Sayajirao University of Baroda	8.74/10
2021	Class-XII (CBSE)	P. P. Savani School, Ankleshwar	94.4%
2019	Class-X (CBSE)	P. P. Savani School, Ankleshwar	97.4%

Work Experience

Research Intern | Student Advance Research Internship Program, IIT Kanpur

Mentor: Dr. Abhishek Kumar Gupta, Department of Electrical Engineering

Project: Detection of Clustering in Stochastic Transmitter Processes via Molecular Communication Signals

Devised a statistical test to detect spatial clustering via a received molecular signal, without knowledge of transmitter locations.

Modelled 3D spatial dynamics via homogeneous and inhomogeneous Poisson Point Processes with decaying intensity.

Derived molecule hitting probabilities via Fick’s diffusion equation, modelling the signal as a sum of Binomial variables.

Simulated the signal across the intensity parameter, extracting peak height, peak time, and rise time as detection statistics.

Built the signal’s theoretical distribution for an ongoing Fisher-Information-based spatial clustering hypothesis test.

(May’26 - Jul’26)

Key Projects

ControlPlane.ai: AI Risk-Control Layer |  | Accenture Innovation Challenge 2026

Building a real-time Python control layer that scores every LLM response for performance, cost, and responsibility risk.

Designing Brier, self-consistency, PSI, and KS monitoring to detect hallucination, miscalibration, and drift.

Implementing a tiered allow/edit/review/block policy with cost-derived thresholds under asymmetric FP/FN costs.

FilingIQ |  | Self Project

Built a RAG pipeline over annual filings of JPMorgan, Bank of America, and Goldman Sachs for financial analysis.

Processed 10,870 text segments, generated all-MiniLM-L6-v2 embeddings, and indexed them in FAISS for retrieval.

Designed a RetrievalQA chain with a Groq-hosted LLM via LangChain, with a Streamlit UI for source attribution.

Vehicle Damage Severity Analysis |  | Self Project

Developed a multi-label deep learning model to classify 6 vehicle damage types using the CarDD image dataset.

Built a computer vision pipeline with COCO annotation parsing, image augmentation, and ImageNet normalization.

Benchmarked custom CNN, VGG16, and ResNet50, achieving the best Macro F1-score of 72.2% with VGG16.

Monetary Policy and Market Volatility |  | Self Project

Analysed the dynamic impact of RBI monetary-policy communication on NIFTY 50 and India VIX using policy documents.

Represented policy content using TF-IDF and GloVe, capturing key themes across inflation, growth, and liquidity.

Combined policy signals with lagged returns and momentum to predict India VIX shocks using RF, RNN, LSTM models.

Yield Curve Forecasting |  | Self Project

Analysed US Treasury yields data alongside macroeconomic indicators to develop a yield-curve forecasting framework.

Evaluated stationarity and temporal dependence using ADF/KPSS and ACF/PACF, comparing ARIMAX, VAR, DNS.

Improved forecast accuracy by up to 32% RMSE over Random Walk and integrated the forecasts into a pricing system.

Scholastic Achievements

Secured All India Rank 5 in Graduate Aptitude Test in Engineering (GATE) Statistics 2025, conducted by IIT Roorkee.

Secured All India Rank 36 in Joint Admission Test for Masters (JAM) Mathematical Statistics 2025, conducted by IIT Delhi.

Granted the National Board Higher Mathematics (NBHM) Scholarship 2025 for pursuing Masters’ Degree in Statistics.

Awarded the Prof. Ashok Manilal Praikh Gold Medal for securing the highest CPI in B.Sc. Statistics (Hons.), 2025.

Qualified for the final round of the AI Hackathon under SARCathon 2025, organized by IIT Bombay.

Cleared 4 actuarial papers: CM1, CS1, CB1, and CB2 from the Institute of Actuaries of India.

Secured AIR-2 in the 13th Statistics Olympiad, 2023, by CR RAO AIMSCS, among participants from India and Sri Lanka.

Position of Responsibility

Secretary, Academics and Career Council, IIT Kanpur

Appointed as the Secretary of the Academics Wing in the AnC council, representing the PG students on academic matters.

Member of the Technical Team for the Institute Research Symposium 2025, managing the Schedule and Abstract Book.

Contributed to the “You Are Not Alone” initiative, relaying anonymous student concerns to institute administration.

(2025-26)

Technical Skills			
Languages	Python Libraries	R Packages	Utilities
Python, R, SQL	NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, Statsmodels	ggplot2, dplyr, shiny, stats	MS Office, L ^A T _E X, R Markdown

Relevant Courses			
Data Science Lab 1 & 2	Non-Linear Regression*	Statistical Inference	Statistical Computing
Time Series Analysis	Game Theory*	Stochastic Processes	Financial Economics